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Selective Strength Loss and Decreased Muscle Activity in Hamstring Injury

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Rehabilitation and prevention strategies for hamstring injuries have focused on muscle strength and flexibility,[5,16,45] with more recent investigations and recommendations suggesting that neuromuscular control[32,49] and sports-specific skills and balance may also be important.[39] Isokinetic dynamometry is commonly used to assess quadriceps and hamstring muscle strength as a potential risk factor for future injury or for residual deficits following hamstring injuries.[6,13,15,16,34,43,49,54,56,57] While some rehabilitation and prevention programs based on strength and flexibility have been shown to reduce the seasonal incidence of hamstring injuries,[15,16,24,57] muscle strength has not consistently been a good indicator of future risk of incurring a hamstring injury[6,43] and of individuals with a hamstring injury.[14,17,63]

STUDY DESIGN: Cross-sectional, controlled laboratory study.

OBJECTIVES: To determine whether thigh muscle isokinetic torque patterns and activity, measured by electromyography (EMG), of individuals with hamstring injury differ from control individuals.

BACKGROUND: Neuromuscular control during thigh muscle strength assessment following hamstring injuries has not been reported.

METHODS: Fifteen athletes with prior hamstring injury (hamstring-injured group [HG]) were compared to 15 uninjured athletes (control group [CG]). The injuries were incurred 6 weeks to 12 months prior to participation, and all injured athletes had returned to at least partial training. Participants performed 5 isokinetic concentric extensor, concentric flexor, and eccentric flexor torque tests at 60°/s in the seated position. Peak torque was determined for each contraction type, as well as average torque for each of 4 time-based movement quartiles. EMG root-mean-squares were calculated in these movement quartiles for the biceps femoris and medial hamstrings.

RESULTS: No significant differences were found for peak torque for all contractions, when comparing HG injured and uninjured sides to CG bilateral averages. The HG injured limb eccentric flexor torque was significantly lower in the fourth quartile (approximately 25° to 5° knee flexion, hamstring lengthened range) compared to the CG bilateral average ($P = .025$). Eccentric flexor biceps femoris and hamstrings EMG root-mean-squares of the HG injured and the uninjured sides were significantly lower in the second to fourth quartiles (towards the lengthened range), compared to the CG bilateral averages ($P<.05$).

CONCLUSION: Decreased strength and EMG activation in a lengthened hamstrings range for the athletes with prior hamstring injury suggested a change in neuromuscular control. Lengthened range assessment of isokinetic eccentric flexor torque may be useful for the assessment of athletes with a prior injury; however, results should be confirmed with prospective studies. *J Orthop Sports Phys Ther 2011;41(5):354-363, Epub 2 February 2011. doi:10.2519/jospt.2011.3268*

KEY WORDS: *dynamometry, EMG, electromyography, strain*

Isokinetic strength variables that have been linked to hamstring injuries include alterations in peak torque (PT), angle of peak torque (APT), and strength ratios. PT is the highest point on the torque curve and represents maximal strength, and the APT indicates the dynamometer's lever position (a proxy for knee angle), at which the PT occurs.[21,51] An alternative term for this variable is "optimal angle for torque,"[7-9] a single measure that better describes the average as opposed to the APT. Although the optimal APT has been used with reference to hamstring injuries,[8,10] low reliability has been reported, particularly for eccentric flexor contractions.[18,46,61]

Isokinetic PT ratios have included the bilateral and the antagonistic ratios. A bilateral or side-to-side ratio compares the measurement recorded during a contraction of a muscle group to that of the contralateral side.[20,21] Two antagonistic PT ratios have been described, which compare the PT of a muscle group to its ipsilateral antagonist. The conventional ratio compares the PT of concentric knee flexor (hamstring) contractions to the PT of ipsilateral concentric knee extensor (quadriceps) contractions. Alternately,

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the functional ratio, or dynamic control ratio, is calculated by comparing the PT of the eccentric knee flexor (hamstring) contraction to the PT of the ipsilateral concentric knee extensor (quadriceps) contraction.[1,17] Although these ratios have been used in research related to athletes with hamstring injuries,[6,43] wide individual variation is found in injured and uninjured populations.[14,15,48] Further, low to moderate reliability has been reported for the antagonistic ratios.[53]

While acting as an agonist during knee flexion, the length-tension relationship and changing moment arm of the hamstrings decrease torque production near full knee extension when the hip joint is in 90° flexion, as in the seated position.[12] Hamstring injuries are most likely to occur during the terminal swing phase of running, while contracting eccentrically and while the muscle is in the more lengthened position.[31] It was suggested in a recent commentary that rehabilitation of athletes with hamstring injuries should progress towards eccentric strengthening exercises for this muscle group in the outer range of motion, that is, where the muscle is in a more lengthened position.[32] In a pilot study conducted in our laboratory on individuals with recent hamstring injuries, residual impairments in isokinetic flexor torque were found for the lengthened range of the hamstrings towards full knee extension.[52] Assessment of muscle strength in this lengthened range may better identify deficits in individuals with recent hamstring injuries.

Decreased hamstring muscle strength following injury is considered a consequence of structural disruption to myofascial tissue.[63] However, it is also influenced by inhibition associated with pain[19] or subconscious fear of pain and recurrence.[42] Discomfort and pain will likely lead to contractile inhibition when testing a previously injured hamstring. The addition of electromyographic (EMG) analysis may potentially help to determine whether neuromuscular sources contribute to the loss of muscle strength following hamstring injuries.

The aims of this study were to determine whether isokinetic torque and the pattern of EMG signal amplitude of the hamstrings would help differentiate individuals who had incurred a hamstring injury from those who have not. It was hypothesized that isokinetic torque and muscle activity of the previously injured hamstring would be significantly reduced during concentric and eccentric flexor contractions in the range towards knee extension (lengthened range), as compared to those of the uninjured side and of a group of control individuals.

METHODS

Participants

Individuals were recruited from the local sports community and physical therapy clinics to participate in this study, which was approved by The University of Otago Human Ethics Committee. All were involved at least twice weekly in sports that included rugby, soccer, hockey, netball, and track athletics. To satisfy inclusion criteria, individuals in the hamstring-injured group (HG) had to have returned to at least partial training. A hamstring injury was defined as a sudden onset of nonimpact posterior thigh pain during a match, competition, or training, within the previous 12 months but not less than 4 weeks previously. The injury also had to be considered of moderate severity,[3] requiring intervention by a health professional, and to have caused the individual to miss at least 1 official match or competition,[6,44] or, alternatively, a minimum of 7 days of regular training. As imaging of hamstring injuries is not routinely performed in community-level athletes[26] or used to define the injury for research purposes,[3,8,62] the history of hamstring injury relied on self-report and a clinical examination by the referring health professional.

Participants in the control group (CG) were recruited from the same sports as the injured group, had to perform full and regular sports training, as well as competition, and could not have a history of a previous hamstring injury. Potential participants were excluded from either group if they reported a previous knee or lumbopelvic injury that required intervention by a health professional or had known neurological, cardiorespiratory, or other systemic disorders.

Isokinetic Dynamometry and Electromyographic Recordings

Isokinetic muscle strength was assessed with a Biodex System 3 Pro isokinetic dynamometer (Biodex Medical Systems, New York, NY). EMG signals of the medial hamstrings and biceps femoris were recorded using an 8-channel TeleMyo telemetric hardware system (Noraxon USA, Inc, Scottsdale, AZ), which included a battery-powered (9 V) transmitter. The analogue EMG signals were bandpass amplified between 16 and 500 Hz, with a gain of 2000, sampled at 1000 Hz, and converted to digital data. The digital data were relayed from the transmitter to the TeleMyo receiver positioned within the line of sight of the transmitter.

Procedures

One week prior to data collection, the participant attended a musculoskeletal screening examination and was familiarized with the isokinetic testing procedures. Flexibility assessments, consisting of the sit-and-reach test[22] and the active knee extension angle,[23] were also performed at this time.

On the day of data collection the EMG signal from each muscle was recorded with pairs of Ag/AgCl surface electrodes (Blue Sensor disposable electrodes, Medicotest, A/S, Denmark), with a center-to-center distance of 25 mm. Surface electrodes were placed on the biceps femoris and the medial hamstrings, according to SENIAM guidelines.[33] The ground electrode was positioned on the sternum. Skin was prepared by shaving and then cleaning with alcohol swabs and abrading firmly with a rough hand towel. Interelectrode impedance was determined for each muscle with a multimeter (Fluke Corporation, Everett,

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TABLE 1 Subject Characteristics*

Variable	CG (n = 15)	HG (n = 15)	P
Age, y	22.5 ± 4.9	24.6 ± 5.1	.212
Anthropometric variables			
Mass, kg	80.0 ± 12.1	81.5 ± 13.0	.718
Height, m	1.82 ± 0.05	1.77 ± 0.06	.014
BMI, kg/m^2	24.0 ± 2.7	26.0 ± 3.8	.078
Flexibility			
Sit-and-reach, cm	28.8 ± 10.3	25.8 ± 7.6	.319
Normalized sit-and-reach, %†	15.9 ± 5.8	14.6 ± 4.5	.466
Active knee extension angle, deg			.814
Preferred (CG) or injured side (HG)	18 ± 11	21 ± 9	
Nonpreferred (CG) or uninjured side (HG)	18 ± 11	20 ± 19	

Abbreviations: BMI, body mass index; CG, control group; HG, hamstring-injured group.
**Values are mean ± SD.*
†Ratio of sit-and-reach measurement to body height.

TABLE 2 Within-Group Comparisons for Peak Torque and Conventional and Functional Ratios With Isokinetic Testing*

Group/Measures	Preferred Side	Nonpreferred Side	Difference†	P
Control group				
Peak torque, Nm/kg				
Concentric extensors	2.92 ± 0.58	2.83 ± 0.60	0.09 (−0.02, 0.19)	.105
Concentric flexors	1.61 ± 0.30	1.60 ± 0.25	0.02 (−0.07, 0.10)	.685
Eccentric flexors	2.60 ± 0.48	2.70 ± 0.50	−0.10 (−0.21, 0.01)	.073
Peak torque ratios, %				
Conventional ratio‡	55.8 ± 7.4	57.7 ± 9.7	−1.8 (−6.2, 2.6)	.387
Functional ratio§	89.9 ± 12.2	97.0 ± 15.7	−7.1 (−12.1, −2.1)	.009
Group/Measures	**Injured Side**	**Noninjured Side**	**Difference†**	**P**
Hamstring-injured group				
Peak torque, Nm/kg				
Concentric extensors	2.56 ± 0.51	2.64 ± 0.55	−0.08 (−0.21, 0.04)	.173
Concentric flexors	1.51 ± 0.42	1.58 ± 0.33	−0.06 (−0.18, 0.05)	.265
Eccentric flexors	2.48 ± 0.64	2.65 ± 0.52	−0.17 (−0.38, 0.05)	.116
Peak torque ratios, %				
Conventional ratio	58.6 ± 10.6	60.3 ± 8.6	−1.7 (−7.1, 3.6)	.496
Functional ratio	96.3 ± 11.5	101.4 ± 13.7	−5.1 (−15.6, 5.3)	.309

**Values are mean ± SD for group descriptions and mean (95% confidence interval) differences between sides within group.*
†Positive values indicate a greater value for the preferred side than the nonpreferred side (control group) or the injured side than the uninjured side (hamstring-injured group).
‡Ratio of concentric hamstring PT to concentric quadriceps PT.
§Ratio of eccentric hamstring PT to concentric quadriceps PT.

WA). The electrodes were removed, and the skin was lightly shaved and cleaned again, if the impedance was greater than 10 kΩ (1% of the system's input impedance). The electrodes and leads were stabilized with adhesive skin tape and self-adherent Handygrip cohesive elastic bandages (BSN Medical Pty Ltd, Mount Waverly, Australia) to decrease movement artifacts.

The order of testing of the injured and uninjured limbs of the HG or the preferred and nonpreferred limbs of the CG was randomized. The participants were assessed in a seated position, and a foam cushion with windows cut into the mid-thigh area was placed on the seat of the dynamometer to decrease thigh pressure on the electrodes placed over the hamstring muscles. The mechanical axis of the dynamometer was aligned with the knee axis of rotation, using the lateral femoral epicondyle as the bony landmark. The total range of motion was 90°, recorded from the anatomically accepted 0° of a fully extended knee to 90° of knee flexion. Prior to the performance of each movement direction, 10 submaximal consecutive concentric and eccentric contractions were performed, followed by 2 maximal contractions as a specific warm-up. Five trials were performed at 60°/s for each contraction direction and type (concentric extensor, concentric flexor, and eccentric flexor contractions). For the concentric extensor contraction, the participants were instructed to extend their knee against the lever's resistance as "hard and fast as possible" from the starting position of 90° knee flexion, followed by passive knee flexion using gravity. The instructions for the concentric flexor contractions were to extend the knee slowly from the starting position of 90° knee flexion, then to "pull down as hard and fast as possible" back to the starting position. For the eccentric flexor contractions, starting with the knee in 90° of flexion, the participants were requested to trigger the lever's movement by first "pushing back" into flexion against the lever, then to resist the movement of knee extension "as hard as possible" until full knee extension, followed by passive knee flexion using gravity. A break of 1 minute was allowed between each of these 3 sets. During the test, all participants were given visual feedback from the monitor and standardized verbal encouragement.

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TABLE 3 Between-Group Differences for Peak Torque and Conventional and Functional Ratios With Isokinetic Testing

	CG Bilateral Average Versus HG Injured Side*		CG Bilateral Average Versus HG Uninjured Side*	
	Mean (95% CI)	*P* Value	Mean (95% CI)	*P* Value
Peak torque, Nm/kg				
Concentric extensors	−0.32 (−0.73, 0.09)	.122	−0.23 (−0.66, 0.19)	.269
Concentric flexors	−0.09 (−0.36, 0.17)	.490	−0.02 (−0.25, 0.20)	.829
Eccentric flexors	−0.17 (−0.59, 0.26)	.433	0.002 (−0.37, 0.38)	.991
Peak torque ratios, %				
Conventional ratio†	1.9 (−5.0, 8.7)	.587	3.6 (−2.5, 9.7)	.237
Functional ratio‡	2.8 (−6.5, 12.1)	.539	8.0 (−2.2, 18.1)	.118

Abbreviations: CG, control group; CI, confidence interval; HG, hamstring-injured group.
**Positive value: HG has a larger value than CG.*
†Ratio of concentric hamstring PT to concentric quadriceps PT.
‡Ratio of eccentric hamstring PT to concentric quadriceps PT.

TABLE 4 Within-Group Interactions (Side by Quartile) for Average Torque and Muscle Activity by Quartiles During Concentric and Eccentric Flexor Contractions*

	HG Injured Side				HG Uninjured Side						
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	*df*†	F	*P*
Concentric flexor contractions											
Average torque‡	84.6 ± 8.8	85.0 ± 5.4	75.1 ± 7.0	55.8 ± 10.3	82.9 ± 10.0	85.3 ± 6.1	75.5 ± 7.7	56.8 ± 11.1	1, 20	0.441	.592
BF EMG RMS	55.6 ± 2.2	54.5 ± 1.9	53.8 ± 2.5	40.3 ± 3.1	54.4 ± 2.7	53.4 ± 2.3	51.1 ± 2.4	42.2 ± 3.5	1, 23	0.761	.468
MH EMG RMS	49.9 ± 11.9	52.0 ± 9.5	58.8 ± 8.2	47.3 ± 13.4	48.0 ± 10.9	48.8 ± 9.2	53.4 ± 12.6	46.1 ± 13.5	1, 24	0.507	.581
Eccentric flexor contractions											
Average torque‡	54.4 ± 9.9	72.6 ± 8.9	84.0 ± 5.6	75.2 ± 13.1	51.6 ± 8.6	68.8 ± 8.0	81.0 ± 5.5	81.2 ± 8.8	1, 17	5.740	.023
BF EMG RMS	55.0 ± 10.6	55.7 ± 10.4	45.0 ± 11.1	28.6 ± 10.9	56.6 ± 11.0	56.4 ± 10.6	50.3 ± 11.1	36.2 ± 10.1	1, 26	1.827	.182
MH EMG RMS	59.2 ± 10.2	51.8 ± 10.1	38.4 ± 11.1	21.5 ± 10.3	57.0 ± 7.2	49.4 ± 11.1	41.3 ± 12.9	27.2 ± 11.3	1, 26	3.531	.055
	CG Preferred Side				**CG Nonpreferred Side**						
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	*df*‡	F	*P*
Concentric flexor contractions											
Average torque‡	84.4 ± 6.7	88.8 ± 2.9	79.2 ± 5.9	58.3 ± 12.4	86.1 ± 4.0	88.5 ± 6.1	78.7 ± 10.1	59.7 ± 14.3	1, 17	0.931	.371
BF EMG RMS	60.8 ± 8.8	55.3 ± 7.4	48.3 ± 11.5	38.2 ± 11.1	61.9 ± 6.8	59.9 ± 7.7	55.5 ± 7.4	37.9 ± 12.7	1, 27	2.850	.077
MH EMG RMS	55.5 ± 9.0	52.7 ± 10.0	50.4± 12.6	45.8 ± 18.1	50.2 ± 12.8	54.1 ± 10.5	55.1 ± 10.8	41.5 ± 12.2	1, 24	2.750	.085
Eccentric flexor contractions											
Average torque‡	46.2 ± 8.7	66.6 ± 8.5	83.4 ± 7.3	85.1 ± 8.6	46.7 ± 9.9	66.3 ± 10.3	81.8 ± 7.8	85.0 ± 11.4	1, 24	0.530	.568
BF EMG RMS	59.3 ± 10.0	63.3 ± 5.9	60.3 ± 6.8	45.4 ± 11.5	53.5 ± 8.5	61.7 ± 6.6	57.5 ± 10.1	45.6 ± 14.7	1, 22	2.341	.129
MH EMG RMS	62.5 ± 8.9	60.5 ± 9.5	53.0 ± 8.2	38.9 ± 10.0	59.0 ± 15.6	57.0 ± 10.7	49.8 ± 11.7	37.1 ± 11.6	1, 18	0.284	.661

Abbreviations: BF, biceps femoris; HG, hamstring-injured group; MH, medial hamstrings; EMG, electromyographic; RMS, root-mean-square; Q, quartile.
**Values are means ± SD percent. Approximate lever angles for quartiles: Q1, 85°-65°; Q2, 65°-45°; Q3, 45°-25°; Q4, 25°-5°.*
†Mauchly's test of sphericity was significant, thus degrees of freedom were adjusted with a Greenhouse-Geisser correction.
‡Torque normalized to peak torque.

EMG data were band-pass filtered between 10 and 450 Hz, using a fourth-order Butterworth filter, and full-wave rectified using LabVIEW Version 5.0 software. Analogue signals of the Biodex dynamometer's lever angular position and velocity were synchronized with the analogue EMG signals and sent to the analogue-to-digital converter using LabVIEW Version 5.0 software. Specifically, isokinetic velocity data were used as the reference marker for processing EMG signals. Filtered EMG data and the digitized lever velocity were saved in a Microsoft Excel spreadsheet.

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Data Analysis

The Biodex software was used to calculate gravity-corrected PT, and the conventional torque ratio (concentric flexor PT/concentric extensor PT) and functional torque ratio (eccentric flexor PT/concentric extensor PT) were calculated for each participant using a Microsoft Excel spreadsheet.

Torque and velocity data, sampled at 200 Hz, were also downloaded from the Biodex software and imported to a Microsoft Excel spreadsheet for analysis of the torque curves, which were drawn for each repetition for the isokinetic movements (ie, periods during which the lever velocity was 60°/s). By excluding data of the nonisokinetic movement, the analysis included torque and EMG data for movement between 85° and 5° knee flexion. The torque curve was smoothed by averaging the torque of each 50-millisecond epoch, resulting in 27 epochs. The peak (without gravity correction) of these curves was determined for each contraction type (concentric extensors, concentric flexors, and eccentric flexors) for normalization of the data.

To assess the torque curve throughout the range of motion, the 27 time epochs for each repetition were grouped into 4 parts (quartiles). Due to the uneven number, the first 3 quartiles had 7 epochs, and the fourth quartile 6. Each quartile corresponded to approximately 20° of lever angular movement. The difference between the fourth quartile and the other 3 was less than 3°. For the concentric extension and the eccentric flexor contractions, the first (approximately 85° to 65° knee flexion) and the fourth (approximately 25° to 5° knee flexion) quartiles represented the movement from the flexed towards the extended knee positions, respectively. For the concentric flexor contractions, the first quartile represented the movement from the extended knee position (approximately 5° to 25° knee flexion) and the fourth quartile towards the final flexed position (approximately 65° to 85° knee flexion). The average torque per movement quartile was normalized to the PT, determined from the Excel analysis of the 5 repetitions.

TABLE 5 Summary of ANOVAs for Between-Group Comparisons (Group by Quartiles) for Normalized Isokinetic Torque and EMG Activity for the HG (n = 15) and the CG (n = 15)

	HG Injured Side Versus CG Bilateral Average			HG Uninjured Side Versus CG Bilateral Average		
Contraction/Measure	***df****	**F**	***P***	***df****	**F**	***P***
Concentric contractions						
Average torque†	2, 55	0.346	.705	2, 24	0.046	.935
Biceps femoris EMG RMS	2, 49	2.113	.136	2, 41	3.960	.069
Medial hamstrings EMG RMS	2, 53	1.508	.213	2, 49	1.219	.300
Eccentric contractions						
Average torque†	1, 32	8.016	.006	1, 35	2.898	.089
Biceps femoris EMG RMS	2, 47	7.248	.003	2, 47	3.170	.059
Medial hamstrings EMG RMS	2, 60	6.450	.002	2, 51	1.746	.187

Abbreviations: ANOVA, analysis of variance; CG, control group; EMG, electromyographic; HG, hamstring-injured group; RMS, root-mean-squares.
**Mauchly's test of sphericity was significant, thus degrees of freedom were adjusted with a Greenhouse-Geisser correction.*
†Average torque normalized to peak torque.

Visual observations from previous research[29] and from our pilot trials have indicated that the knee angle and dynamometer lever angle may differ up to 20°. Further, low test-retest reliability has been reported for the lever's angular position.[18,46,61] The quartiles were thus calculated on the basis of time, rather than dynamometer lever angle.

The EMG root-mean-squares (RMS) were also averaged to 50-millisecond epochs, with reference to the start of isokinetic movement, and normalized to the peak EMG RMS of each movement direction and contraction type (flexion/extension and eccentric/concentric). The normalized data of each repetition were divided into 4 movement quartiles, as described for the isokinetic data. This was an adaptation of the technique recommended by Kellis and Baltzopoulos.[38] Instead of dividing the movement into epochs of 10°,[38] a larger range of approximately 20° was chosen. The test-retest reliability for the concentric and eccentric torque quartiles was determined with 20 uninjured individuals (mean ± SD age, 21 ± 3 years), with a time interval between test-retest of 1 week. Intraclass correlation coefficients ($ICC_{3,1}$) were between 0.80 and 0.93 for the eccentric flexor quartiles, and between 0.77 and 0.88 for the concentric flexor quartiles.

Statistical Analysis

All statistical analyses were conducted with SPSS 14.0 software (SPSS Inc, Chicago, IL). Descriptive statistics were computed for all variables. The analysis was conducted in 2 stages. First, a within-group (HG and CG) analysis was conducted to explore differences between the injured and noninjured limbs and the preferred and nonpreferred limbs (bilateral differences). Because no differences between sides were detected in the noninjured group, the averages of both sides were used for subsequent between-group comparisons. Second, between-group analyses were performed, comparing the HG injured side to the CG bilateral average, and the HG uninjured side to the CG bilateral average.

Within-group differences for PT and the PT ratios were analyzed using paired *t* tests. Independent *t* tests were used to

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TABLE 6 Post Hoc Between-Group Comparisons for Isokinetic Torque and EMG Signal Amplitude for the Eccentric Quartiles for the HG (n = 15) and the CG (n = 15)*

	HG Injured Side Versus CG Bilateral Average†	
Variable	**Difference**	***P***
Average torque, %		
Q1	8.0 (1.0, 14.9)	.026
Q2	6.2 (−0.3, 12.6)	.061
Q3	1.4 (−3.1, 5.9)	.537
Q4	−9.9 (−18.4, −1.3)	.025
BF EMG RMS, %‡		
Q1	−2.0 (−9.1, 5.1)	.570
Q2	−6.8 (−12.8, −0.7)	.030
Q3	−14.3 (−21.3, −7.3)	<.001
Q4	−17.7 (−26.8, −8.6)	<.001
MH EMG RMS, %		
Q1	−2.3 (−10.6, 6.1)	.586
Q2	−7.7 (−14.8, −0.7)	.032
Q3	−13.6 (−21.2, −6.0)	.001
Q4	−16.8 (−24.3, −9.4)	<.001

Abbreviations: BF, biceps femoris; CG, control group; EMG, electromyographic; HG, hamstring-injured group; MH, medial hamstrings; Q, quartile; RMS, root-mean-square.
**Data are mean (95% confidence interval) differences. Approximate lever angles for quartiles: Q1, 85°-65°; Q2, 65°-45°; Q3, 45°-25°; Q4, 25°-5°.*
†Positive value: HG has a larger value than CG.
‡n = 14 for biceps femoris EMG RMS of HG injured side.

explore differences for these variables between the HG injured and uninjured sides, and the CG bilateral average.

For average torque and EMG RMS quartiles, the within-group analyses were conducted with 2-by-4 (sides by quartiles) repeated-measures analyses of variance (ANOVAs). Between-group analyses were performed with 2-by-4 (group-by-quartiles) mixed-model ANOVAs. Significant differences were explored using a post hoc Bonferroni procedure to control for type I error, and the a priori level of statistical significance was set at $P<.05$

RESULTS

Participants

Fifteen male control participants and 15 males with recent hamstring injuries (**TABLE 1**) volunteered and gave written informed consent to participate in this study. The mean ± SD time since injury for the HG was 3.6 ± 3.5 months (range, 1-12 months), and the time before a return to partial training had been 3.9 ± 1.9 weeks. Six of the participants had missed up to 3 weeks of full training due to the injury, with 9 needing 4 or more weeks. Based on self-report, 7 (47%) had incurred their hamstring injury on the nonpreferred limb, and 9 (60%) had incurred the injury in the lateral hamstrings. The injury had been localized to the mid-thigh in 9 (60%), proximal hamstrings in 5 (33%), and distal hamstrings in 1 (7%) of the participants. Eight participants reported that they had previous hamstring injuries: 3 in the same limb, 3 in the opposite limb, and 2 bilaterally. The time of these injuries ranged from 6 months to 3 years prior to the current injury.

At the time of testing, all participants with recent hamstring injuries had returned to sports training, but only 3 of these had returned to preinjury competitive and training levels. Of the 12 participants who had not returned to preinjury competitive training levels, 10 had returned to sports activities, indicating that they had only mild limitations with respect to strenuous sports or moderate work. The remaining 2 participants reported discomfort in the hamstrings during moderate work or sports activities. Three HG participants were still undergoing exercise-based rehabilitation supervised by a health professional at the time of testing.

Peak Torque Variables

There were no significant bilateral differences within the CG and the HG for PT and the PT ratios, with the exception for the functional PT ratio within the CG (**TABLE 2**). This ratio was lower in the CG preferred leg than the nonpreferred leg ($P = .009$); however, the difference was less than the previously reported standard error of measurement (7.65%).[53] There were no statistically significant differences for the PT variables and the conventional and functional PT ratios when comparing HG injured and uninjured sides to the CG bilateral averages (**TABLE 3**).

Average Torque and EMG RMS for Concentric Flexor Contractions

The biceps femoris electrodes were dislodged during data collection on the injured side of 1 HG participant, resulting in data for 14 HG participants for this muscle. No significant side-by-quartile interactions were found for within-group differences for average torque and EMG RMS; thus the injured and uninjured sides of the HG and the preferred and nonpreferred sides of the CG showed no difference in performance between quartiles (**TABLE 4**). Significant main effects were found for the quartiles but not for sides. For between-group comparisons of the concentric flexor torque and for EMG RMS, no significant group-by-quartile interactions were found, when comparing the HG injured and uninjured sides,

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respectively, to the CG bilateral average (**TABLE 5**).

Average Torque and EMG RMS for Eccentric Flexor Contractions

For eccentric flexor contractions, a side-by-quartile interaction was found for average torque when comparing the HG injured side to the uninjured side (P = .023) (**TABLE 4**). However, post hoc testing found no statistically significant differences when comparing the individual torque quartiles between injured and uninjured sides (P>.05). No significant side-by-quartile interactions were found for the preferred and nonpreferred sides of the CG for eccentric torque (**TABLE 4**).

For the between-group comparison for eccentric flexor torque and EMG RMS, significant interactions were found for group by quartile, when comparing the variables for the HG injured side to the CG bilateral average (**TABLE 5**). No interactions were found for group by quartile when comparing the torque and EMG RMS quartiles of the HG uninjured side to the CG bilateral average (**TABLE 5**).

Post hoc tests comparing the respective torque quartiles of the HG injured side to the CG bilateral average showed significant differences for quartile 1 (approximately 85° to 65° knee flexion; P = .026) and quartile 4 (approximately 25° to 5° knee flexion; P = .025) (**TABLE 6**, **FIGURE 1**). Significantly lower biceps femoris (**FIGURE 2**) and the medial hamstrings EMG RMS from quartile 2 (approximately 65° to 45°) to quartile 4 (approximately 5° to 25°) were found when comparing the HG injured and the HG uninjured sides, respectively, to the CG bilateral average (**TABLE 6**).

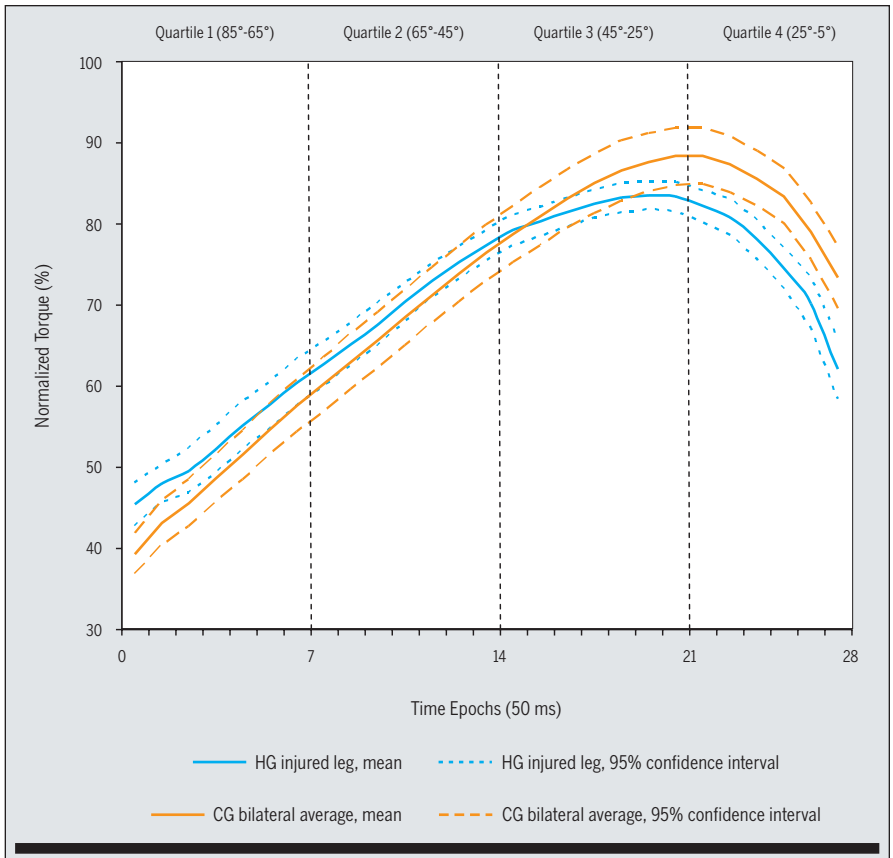

FIGURE 1. Eccentric flexor torque of individuals with a recent hamstring injury (HG) and the bilateral average of the control group (CG). Data are mean and 95% confidence intervals. The torque was normalized to the peak torque.

DISCUSSION

To our knowledge, this is the first published observation of a selective decrease in isokinetic eccentric flexor torque of hamstring-injured limbs in the lengthened range of contraction, despite our having found no significant between-group differences for concentric and eccentric flexor PT and for the PT ratios. The findings of no significant differences for PT and the PT ratios between injured and control participants are consistent with a number of previous cross-sectional studies based on isokinetic assessments[8,63] but contrast with those of others.[15,36,45] In the current study, all HG participants had returned to sports training, with only 2 experiencing residual symptoms during moderate sporting activities. The relatively high functional level of these participants may explain the lack of significant between-group PT differences.

During eccentric flexor contractions, the average torque of quartile 1 (approximately 85° to 65°) was significantly higher in the HG injured limb than the CG bilateral average but lower for quartile 4 (approximately 25° to 5° lengthened range). Loss of muscle strength following muscle injury has been attributed to loss of structural integrity, such as changes to sarcomeres,[8,41,58] fatty infiltration, and changes in muscle volume.[50] Evidence of long-term (5 to 23 months postinjury) musculotendon remodeling was recently demonstrated using magnetic resonance imaging in athletes who had returned to sport for at least 1 month prior to the study, following a clinically diagnosed hamstring injury.[50] The changes in the eccentric flexor torque curve of the hamstring-injured limb may be associated with loss of flexibility or increased stiffness.[28] However, the sit-and-reach and active knee extension angles were not different between the HG and CG (**TABLE 1**), indicating that these clinical measures of flexibility were unlikely to explain the between-group torque differences. Assessing resistance to passive lengthening of the hamstring muscles[27] would have been needed to fully exclude a potential role of increasing hamstring stiffness to

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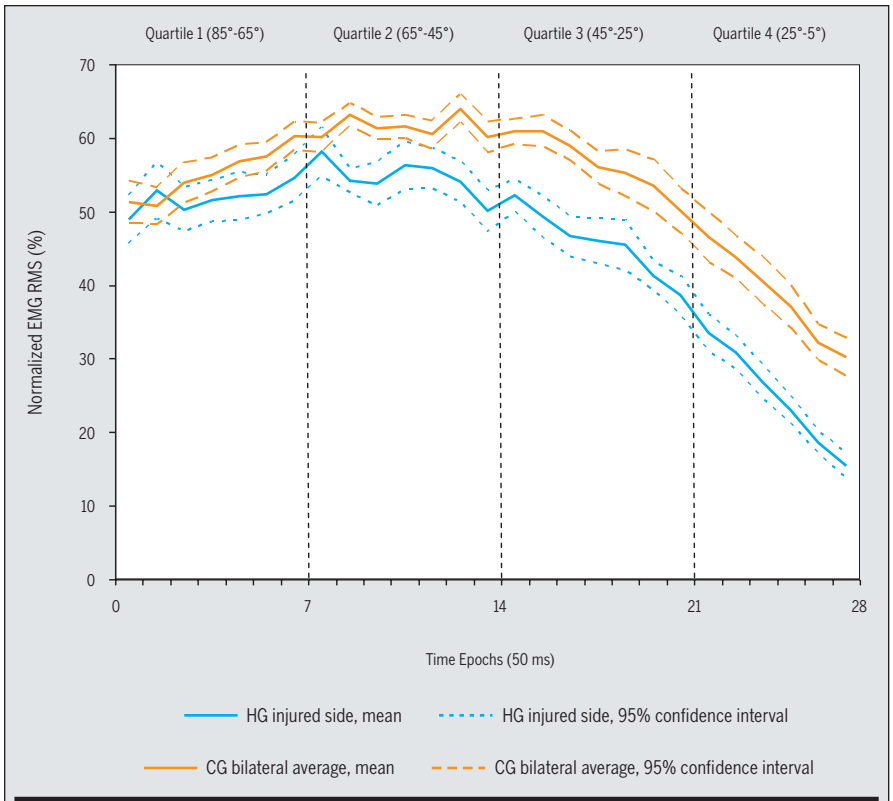

FIGURE 2. Normalized EMG RMS of biceps femoris during eccentric flexor contractions of individuals with a recent hamstring injury (HG) and the bilateral average of the control group (CG). Data are mean and 95% confidence intervals.

explain the between-group differences in eccentric flexor torque. Although magnetic resonance imaging was not used in the current study to confirm presence of scar tissue, long-term structural changes might explain, at least partly, the changes in the eccentric torque curve.

Additionally, the between-group differences in EMG amplitudes during eccentric flexor contractions indicate that neurophysiological factors need to be considered. For participants of the CG, normalized EMG amplitudes decreased and eccentric flexor average torque increased from the start to the lengthened range of the muscles. These findings are similar to those found in uninjured participants during isometric flexor contractions in different knee flexion positions.[40]

Strength assessments that require maximal effort are affected by the presence of pain,[19] and/or the apprehension of pain,[47] most likely mediated by central inhibitory neurophysiological mechanisms,[55] as well as peripheral changes.[58] It is particularly during maximal eccentric contractions that neural inhibitory mechanisms are believed to be activated.[40] Further, pain, discomfort, or a decreased tolerance for discomfort following muscle injury is expected when the structure is stretched or lengthened. In the current study, participants of the HG had already returned to sports training and only 3 were still undergoing rehabilitation. Despite having found no statistically significant between-group differences for PT for all contractions, we did find a significant difference for outer-range eccentric torque. Tendon and muscle remodeling may persist for many months following a hamstring injury,[50] and scarring could contribute to the decreased torque in the outer range of motion and potentially to decreased outer-range torque. It is also possible that the individual may not be willing to exert a maximal contraction in the range of motion in which the muscle is perceived to be "at risk," and that the reduced ability to contract maximally may persist even after symptoms during functional activities have resolved.[55] The findings of our study support clinical recommendations[32] that strengthening exercises in the final phase of rehabilitation of patients with hamstring injuries should include eccentric strengthening performed towards the end range of motion (lengthened range).

It was interesting to note that the difference between the biceps femoris RMS EMG of the HG uninjured limb and that of the CG had a significance level of $P = .059$ (**TABLE 5**). This might have been due to 3 of the participants having incurred a previous hamstring injury on that side. Alternatively, central inhibitory neurophysiological mechanisms[55] might have been present.

In the current study, the localization of the injury was based on self-report and on the clinical diagnosis of the referring healthcare professional. Although self-report of injury location has been shown to be reliable within 12 months of incurring the injury,[25] information with regards to severity of the initial injury and exact diagnosis could not be confirmed. However, in New Zealand, physical therapists are trained to be competent in assessing muscle injuries as first-line practitioners. The individuals of the HG were representative of those generally assessed and treated in community clinics for hamstring injuries, as defined by the inclusion criteria. Further, 8 of the injured participants reported having a previous hamstring injury. Future studies should explore whether findings of this study are more likely to be present in athletes with recurrent injuries than in those with first-time injuries.

Two further limitations need to be highlighted. First, this design did not allow clear inference of causality.[2] Therefore, this research does not provide evidence for changes in muscle activity or isokinetic variables as contributing

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factors in hamstring injury or reinjury. Further, the study design does not allow inferences for determining return to sport. Conclusions are limited to establishing whether there is evidence for within-group and between-group differences for the dependent variables. Our findings should be followed by a prospective study to confirm the utility of the assessment of eccentric flexor torque towards the lengthened position for predicting the risk of incurring hamstring injury or reinjury. Further, the testing velocity was limited to 60°/s. The velocity of 60°/s has also been used in previous studies relating to hamstring injuries[4,8,56] and in studies investigating hamstring muscle activity during isokinetic testing.[30,35,37] However, other velocities have also been used.[16] Performing eccentric contractions at different speeds is unlikely to have significant effects on torque and muscle activity.[11,59,60] It is thus possible that the addition of slower or faster speeds for the eccentric contractions would not have changed our conclusions. However, future studies could investigate isokinetic torque and EMG patterns at different speeds.

CONCLUSION

It is possible that both structural as well as neurophysiological factors contributed to decreased torque and agonist hamstring muscle activity in the lengthened range of eccentric flexor contractions in participants with a recent history of a hamstring injury. ◉

KEY POINTS

FINDINGS: Eccentric flexor torque was found to be lower towards the lengthened range in the hamstring-injured limbs in comparison to uninjured limbs, despite no significant differences for PT variables between control and hamstring-injured groups.

IMPLICATION: Final-phase rehabilitation of athletes with hamstring injuries may require eccentric muscle strengthening towards the lengthened position. Eccentric flexor torque performance in the lengthened flexor range might be a useful indicator for determining capacity to return to training following an injury.

CAUTION: This was a cross-sectional study design and findings need to be confirmed with a prospective study.

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